

OOP-TASK

Lab-07

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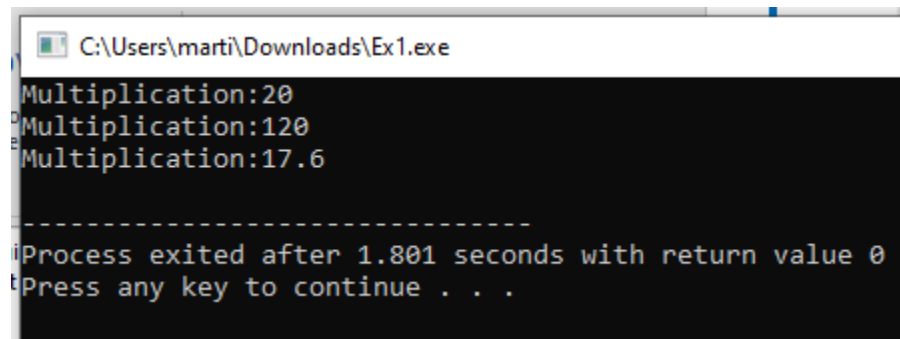
Rollno: **25K-0868**

Exercises

Q1)Ex1:

```
1 // Create a class Calculator that overloads a function multiply() to perform:
2
3 // Multiplication of two integers
4 // Multiplication of three integers
5 // Multiplication of two floating values
6 #include <iostream>
7 using namespace std;
8 class Calculator{
9     public:
10     void Multiply(int a,int b){
11         cout << "Multiplication:" << a*b << endl;
12     }
13     void Multiply(int a,int b,int c){
14         cout << "Multiplication:" << a*b*c << endl;
15     }
16     void Multiply(float a,float b){
17         cout << "Multiplication:" << a*b << endl;
18     }
19 };
20 int main(){
21     Calculator c;
22     c.Multiply(4,5);
23     c.Multiply(4,5,6);
24     c.Multiply(3.2f,5.5f);
25     return 0;
26 }
```

Output:



```
C:\Users\marti\Downloads\Ex1.exe
Multiplication:20
Multiplication:120
Multiplication:17.6
-----
Process exited after 1.801 seconds with return value 0
Press any key to continue . . .
```

Ex-2:

```
10  #include <iostream>
11  using namespace std;
12
13  class Shape {
14  public:
15      virtual void draw() {
16          cout << "Drawing a generic shape" << endl;
17      }
18
19      virtual ~Shape() {}
20  };
21
22  class Circle : public Shape {
23  public:
24      void draw() override {
25          cout << "Drawing Circle" << endl;
26      }
27  };
28
29  class Rectangle : public Shape {
30  public:
31      void draw() override {
32          cout << "Drawing Rectangle" << endl;
33      }
34  };
35
36  int main() {
37      Circle c1;
38      Rectangle r1;
39
40      // Base-class pointers call the overridden functions at runtime.
41      Shape* shapes[2] = {&c1, &r1};
42
43      for (int i = 0; i < 2; i++) {
44          shapes[i]->draw();
45      }
46
47      return 0;
48  }
```

Output:

C:\Users\marti\Downloads\Ex1.exe

Drawing Circle

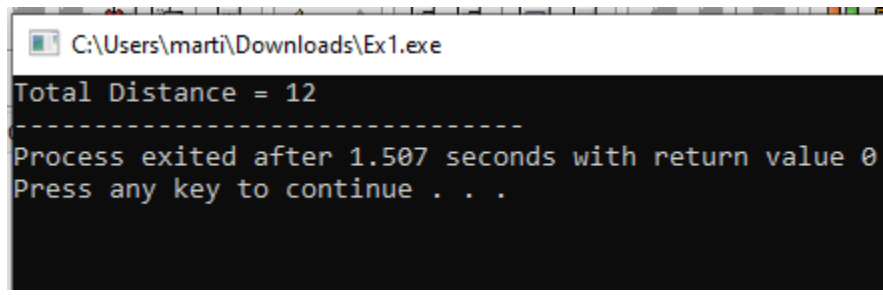
Drawing Rectangle

Process exited after 2.167 seconds with return value 0
Press any key to continue . . .

Ex-3:

```
1  #include <iostream>
2  using namespace std;
3
4  class Distance
5  {
6  public:
7
8      int meters;
9
10     Distance(int m)
11     {
12         meters = m;
13     }
14
15     Distance operator + (Distance d)
16     {
17         return Distance(meters + d.meters);
18     }
19 };
20
21 int main()
22 {
23     Distance d1(5);
24     Distance d2(7);
25
26     Distance d3 = d1 + d2; //d3=d1.operator+(d2)
27
28     cout << "Total Distance = " << d3.meters;
29
30     return 0;
31 }
```

Output:

A screenshot of a Windows command prompt window. The title bar at the top reads "C:\Users\marti\Downloads\Ex1.exe". The command prompt area has a black background with white text. The output displayed is: "Total Distance = 12", followed by a dashed line "-----", then "Process exited after 1.507 seconds with return value 0", and finally "Press any key to continue . . .".

```
C:\Users\marti\Downloads\Ex1.exe
Total Distance = 12
-----
Process exited after 1.507 seconds with return value 0
Press any key to continue . . .
```

TASKS

Task-1:

```
1  #include <iostream>
2  using namespace std;
3  #define PI 3.142
4  class Shape{
5  public:
6  void area(int side){
7      cout << "Area of square is: " << side * side << endl;
8  }
9  void area(int length,int width){
10     cout << "Area of rectangle is: " << length * width << endl;
11 }
12 void area(double radius){
13     cout << "Area of circle is: " << PI * radius * radius << endl;
14 }
15 };
16 int main(){
17     Shape shape;
18
19     int side, length, width;
20     double radius;
21
22     cout << "Enter side of square: ";
23     cin >> side;
24     shape.area(side);
25
26     cout << "Enter length and width of rectangle: ";
27     cin >> length >> width;
28     shape.area(length, width);
29
30     cout << "Enter radius of circle: ";
31     cin >> radius;
32     shape.area(radius);
33
34     return 0;
35 }
```

 C:\Users\marti\Downloads\Ex1.exe

```
Enter side of square: 3
Area of square is: 9
Enter length and width of rectangle: 12
3
Area of rectangle is: 36
Enter radius of circle: 5
Area of circle is: 78.55

-----
Process exited after 6.745 seconds with return value 0
Press any key to continue . . . █
```

Task-2:

```

12 class Manager : public Employee {
13 public:
14     void calculateSalary() override {
15         cout << "Manager salary: 20000" << endl;
16     }
17 };
18
19 class Developer : public Employee {
20 public:
21     void calculateSalary() override {
22         cout << "Developer salary: 15000" << endl;
23     }
24 };
25
26 class Intern : public Employee {
27 public:
28     void calculateSalary() override {
29         cout << "Intern salary: 5000" << endl;
30     }
31 };
32
33 int main() {
34     Employee* employees[3];
35
36     Manager manager;
37     Developer developer;
38     Intern intern;
39
40     employees[0] = &manager;
41     employees[1] = &developer;
42     employees[2] = &intern;
43
44     for (int i = 0; i < 3; i++) {
45         employees[i]->calculateSalary();
46     }
47
48     return 0;
49 }

```


C:\Users\marti\Downloads\Ex1.exe

```
Manager salary: 20000
Developer salary: 15000
Intern salary: 5000

-----
Process exited after 1.197 seconds with return value 0
Press any key to continue . . .
```

Task-3:

```
1  #include <iostream>
2  using namespace std;
3
4  class Complex {
5  private:
6      double real;
7      double imag;
8
9  public:
10     Complex(double r = 0, double i = 0) {
11         real = r;
12         imag = i;
13     }
14
15     Complex operator+(const Complex& other) const {
16         return Complex(real + other.real, imag + other.imag);
17     }
18
19     Complex operator-(const Complex& other) const {
20         return Complex(real - other.real, imag - other.imag);
21     }
22
23     Complex operator*(const Complex& other) const {
24         return Complex(
25             (real * other.real) - (imag * other.imag),
26             (real * other.imag) + (imag * other.real)
27         );
28     }
29
30     void display() const {
31         cout << real;
32         if (imag >= 0) {
33             cout << " + " << imag << "i";
34         } else {
35             cout << " - " << -imag << "i";
36         }
37     }
38 };
39
```

```

39
40 int main() {
41     Complex c1(4, 3);
42     Complex c2(2, -5);
43
44     Complex sum = c1 + c2;
45     Complex diff = c1 - c2;
46     Complex prod = c1 * c2;
47
48     cout << "c1 = ";
49     c1.display();
50     cout << endl;
51
52     cout << "c2 = ";
53     c2.display();
54     cout << endl;
55
56     cout << "c1 + c2 = ";
57     sum.display();
58     cout << endl;
59
60     cout << "c1 - c2 = ";
61     diff.display();
62     cout << endl;
63
64     cout << "c1 * c2 = ";
65     prod.display();
66     cout << endl;
67
68     return 0;
69 }

```

C:\Users\marti\Downloads\Ex1.exe

```

c1 = 4 + 3i
c2 = 2 - 5i
c1 + c2 = 6 - 2i
c1 - c2 = 2 + 8i
c1 * c2 = 23 - 14i

```

```

-----
Process exited after 1.549 seconds with return value 0
Press any key to continue . . .

```

Task-4:

```
1  #include <iostream>
2  using namespace std;
3
4  class Vehicle {
5  public:
6      virtual void startEngine() {
7          cout << "Vehicle engine started." << endl;
8      }
9
10     virtual ~Vehicle() {}
11 };
12
13 class Car : public Vehicle {
14 public:
15     void startEngine() override {
16         cout << "Car engine started with key ignition." << endl;
17     }
18 };
19
20 class Bike : public Vehicle {
21 public:
22     void startEngine() override {
23         cout << "Bike engine started with self-start button." << endl;
24     }
25 };
26
27 class Truck : public Vehicle {
28 public:
29     void startEngine() override {
30         cout << "Truck engine started with heavy-duty ignition." << endl;
31     }
32 };
33
34 int main() {
35     Car car;
36     Bike bike;
37     Truck truck;
38
39     Vehicle* vehicles[3] = {&car, &bike, &truck};
40
41     for (int i = 0; i < 3; i++) {
42         vehicles[i]->startEngine();
43     }
44
45     return 0;
46 }
```

```
C:\Users\marti\Downloads\Ex1.exe
Car engine started with key ignition.
Bike engine started with self-start button.
Truck engine started with heavy-duty ignition.

-----
Process exited after 1.574 seconds with return value 0
Press any key to continue . . .
```

ADDITIONAL PRACTICE EXERCISES:

Ex-1:

```
1  #include <iostream>
2  using namespace std;
3
4  class Profile{
5      private:
6          string name;
7          int age;
8          string status;
9      public:
10         Profile(string name,int age,string status):name(name),age(age),status(status){}
11         friend ostream& operator<<(ostream &out,const Profile& obj);
12     };
13     ostream& operator<<(ostream &out,const Profile& obj){
14         out << "Profile details" << endl;
15         out << "Name :" << obj.name << endl;
16         out << "Age :" << obj.age << endl;
17         out << "Status :" << obj.status << endl;
18         return out;
19     }
20     int main(){
21         Profile p("Abdullah",19,"student");
22         cout << p ;
23     }
```

```
C:\Users\marti\Downloads\Exercise1.exe
Profile details
Name :Abdullah
Age :19
Status :student

-----
Process exited after 1.619 seconds with return value 0
Press any key to continue . . .
```

Ex-2:

```
2 using namespace std;
3
4 #define row 3
5 #define col 3
6 class Matrix{
7     private:
8         int Mat[row][col];
9     public:
10        Matrix(){}
11        Matrix(int **mat){
12            for(int i = 0; i < row; i++){
13                for(int j = 0; j < col; j++){
14                    Mat[i][j] = mat[i][j];
15                }
16            }
17        }
18        friend ostream& operator<<(ostream & out, const Matrix& M){
19            for(int i = 0; i < row; i++){
20                for(int j = 0; j < col; j++){
21                    out << M.Mat[i][j] << " ";
22                }
23                out << "\n";
24            }
25            out << "\n";
26            return out;
27        }
28        Matrix operator+(Matrix m){
29            Matrix tempM;
30            for(int i=0;i<row;i++){
31                for(int j=0;j<col;j++){
32                    tempM.Mat[i][j] = this->Mat[i][j] + m.Mat[i][j];
33                }
34            }
35            return tempM;
36        }
37    };
38    int main(){
39        int arr1[3][3] = {{1,2,3},{1,2,3},{1,2,3}};
40        int arr2[3][3] = {{1,2,3},{1,2,3},{1,2,3}};
41        int *Mat1[3] = {arr1[0], arr1[1], arr1[2]};
42        int *Mat2[3] = {arr2[0], arr2[1], arr2[2]};
43        Matrix m1((int**)Mat1);
44        Matrix m2((int**)Mat2);
45        Matrix m3 = m1 + m2;
46        cout << m3;
47        return 0;
48    }
```

```
C:\Users\marti\Downloads\Exercise1.exe
2 4 6
2 4 6
2 4 6

-----
Process exited after 1.88 seconds with return value 0
Press any key to continue . . .
```

Ex-3:

```
1  #include <iostream>
2  using namespace std;
3  class Shape{
4      public:
5          virtual void area(){
6              cout << "default area\n";
7          }
8  };
9  class Circle:public Shape{
10     private:
11         int radius;
12     public:
13         Circle(int r):radius(r){}
14         void area() override{
15             cout << "circle area " << 3.142*radius*radius<< endl;
16         }
17 };
18 class Rectangle:public Shape{
19     private:
20         int x;
21         int y;
22     public:
23         Rectangle(int x,int y):x(x),y(y){}
24         void area() override{
25             cout << "Rectangle area " << x*y << endl;
26         }
27 };
28
29
30 int main(){
31     Shape *s;
32     Circle c(5);
33     s=&c;
34     s->area();
35     Rectangle r(2,3);
36     s = &r;
37     s->area();
38     return 0;
39 }
```

 C:\Users\marti\Downloads\Exercise1.exe

circle area 78.55

Rectangle area 6

Process exited after 1.068 seconds with return value 0

Press any key to continue . . .